

NKPC HSA Bayesian State-Space Estimation: Output gap, HP filter

July 9, 2026

1 Data and Transformations

Raw data are read from `data/raw/`. Processed model-ready data are written to `data/processed/`. The HSA competition aggregator convention is

$$N_t^{model} = \frac{100 \log(N_t^{level}) - \overline{100 \log(N^{level})}}{10}.$$

Thus N_t^{model} , $\hat{N}_t$, and $\bar{N}_t$ are centered and measured in ten-log-point units. If a run metadata file records another `n_transform`, interpret the corresponding coefficients using that metadata.

2 Model Specifications

The CES benchmark is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + \kappa x_t + e_t.$$

The dynamic HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + \kappa x_t - \theta \hat{N}_t + e_t.$$

The steady HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + (\kappa_0 + \delta \bar{N}_t) x_t + e_t.$$

The both-effects HSA specification is

$$\pi_t = \alpha \pi_{t-1} + (1 - \alpha) E_t \pi_{t+1} + (\kappa_0 + \delta \bar{N}_t) x_t - (\theta_0 + \gamma \bar{N}_t) \hat{N}_t + e_t.$$

The maintained HSA decomposition is

$$N_t = \hat{N}_t + \bar{N}_t, \quad \hat{N}_t = \rho_1 \hat{N}_{t-1} + \rho_2 \hat{N}_{t-2} + u_t, \quad \bar{N}_t = n + \bar{N}_{t-1} + \epsilon_t.$$

3 Unit and Covariance Conventions

Priors for κ , κ_0 , and δ are specified in physical units. Legacy Gibbs samplers internally use `KAPPA_SCALE=100` when the design matrix contains $x_t/100$. All posterior draws, tables, figures, SDDR calculations, and model-comparison summaries are reported in physical units. The default dynamic-HSA shock covariance allows only (e_t, ζ_t) correlation; diagonal and historical full covariance modes are available only through explicit run metadata. Because the default N transform is centered and divided by 10, the reported δ , θ , θ_0 , and γ coefficients are already effects for a ten-log-point movement in the corresponding N component. κ_0 and θ_0 are intercepts at average $\bar{N}_t$.

4 Coefficient Summary

Table 1 reports posterior means and 95% credible intervals for key parameters across all four models (baseline prior, full sample). Cells show *mean* [*ci_2.5*, *ci_97.5*].

Table 1: Posterior means and 95% credible intervals by model

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.670 [0.566, 0.773]	0.665 [0.559, 0.770]	0.668 [0.567, 0.769]	0.669 [0.561, 0.779]
kappa	0.138 [0.047, 0.227]	NaN	0.143 [0.055, 0.230]	NaN
kappa_0	NaN	0.140 [0.049, 0.228]	NaN	0.140 [0.048, 0.230]
delta	NaN	-0.005 [-0.038, 0.027]	NaN	-0.005 [-0.038, 0.027]
theta	NaN	NaN	0.054 [-0.157, 0.328]	NaN
theta_0	NaN	NaN	NaN	0.049 [-0.141, 0.263]
gamma	NaN	NaN	NaN	-0.001 [-0.039, 0.036]
rho_1	NaN	0.279 [0.092, 0.463]	0.296 [0.016, 0.658]	0.279 [-0.122, 0.441]
rho_2	NaN	-0.844 [-0.967, -0.380]	-0.757 [-0.957, -0.200]	-0.852 [-0.963, -0.442]
phi_1	0.896 [0.821, 0.972]	0.895 [0.823, 0.970]	0.898 [0.826, 0.966]	0.897 [0.820, 0.969]
lambda_ez	0.134 [-0.093, 0.361]	0.130 [-0.106, 0.371]	NaN	0.133 [-0.100, 0.360]
n	NaN	-0.033 [-0.064, 0.000]	-0.030 [-0.062, 0.006]	-0.031 [-0.060, 0.001]

5 Result Blocks

Results are organized by condition. Each section fixes a data specification, competition measurement frequency, prior specification, and sample period, then reports the coefficient table, prior/posterior overlay by model, Savage-Dickey density ratios, time-varying κ_t path, and model-comparison output for that condition.

6 output_gap_hp: quarterly_interpolated, prior=baseline, period=full

Condition. data=output_gap_hp, competition=quarterly_interpolated, prior=baseline, period=full, constraint=unrestricted (unrestricted).

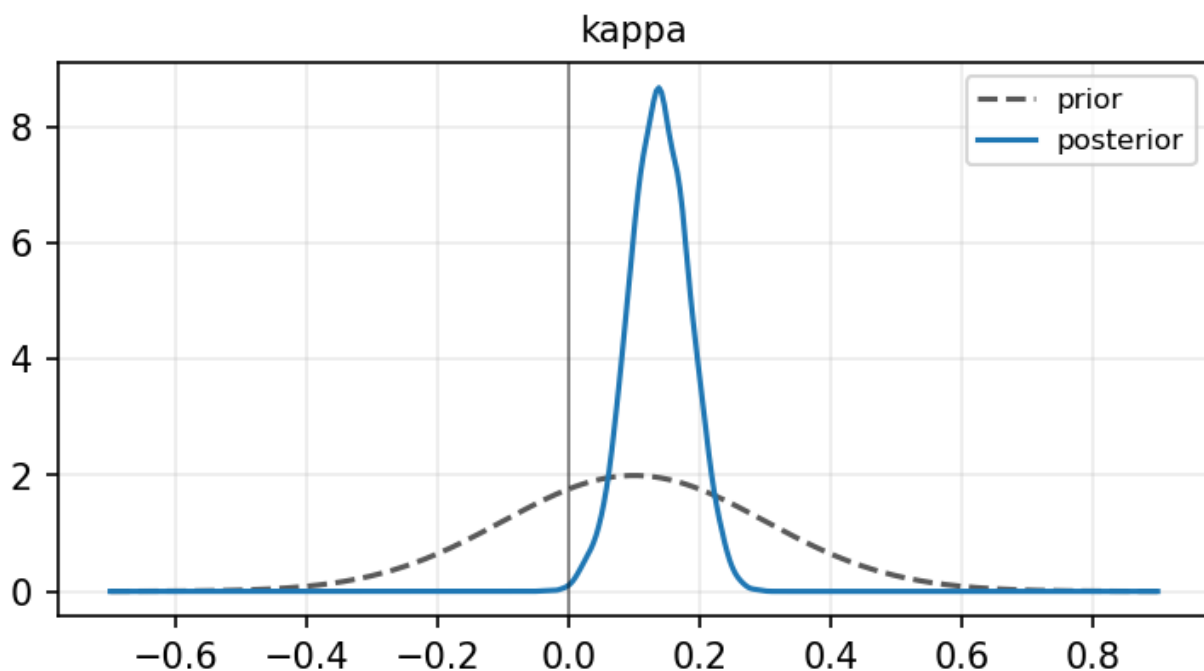
6.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.670 [0.566, 0.773]	0.666 [0.564, 0.768]	0.662 [0.555, 0.763]	0.660 [0.548, 0.764]
kappa	0.138 [0.047, 0.227]	NaN	0.137 [0.042, 0.232]	NaN
kappa_0	NaN	0.142 [0.048, 0.233]	NaN	0.136 [0.040, 0.229]
delta	NaN	-0.002 [-0.031, 0.026]	NaN	-0.006 [-0.037, 0.026]
theta	NaN	NaN	-0.017 [-0.122, 0.123]	NaN
theta_0	NaN	NaN	NaN	0.022 [-0.091, 0.135]
gamma	NaN	NaN	NaN	-0.021 [-0.057, 0.017]
rho_1	NaN	1.805 [1.703, 1.896]	1.781 [1.667, 1.886]	1.798 [1.687, 1.891]
rho_2	NaN	-0.818 [-0.905, -0.721]	-0.794 [-0.894, -0.685]	-0.813 [-0.903, -0.705]
phi_1	0.896 [0.821, 0.972]	0.898 [0.822, 0.973]	0.898 [0.824, 0.968]	0.896 [0.821, 0.969]
lambda_ez	0.134 [-0.093, 0.361]	0.127 [-0.100, 0.356]	NaN	0.132 [-0.090, 0.357]
n	NaN	-0.044 [-0.063, -0.020]	-0.044 [-0.069, 0.001]	-0.039 [-0.050, -0.029]

6.2 Prior vs Posterior by Model

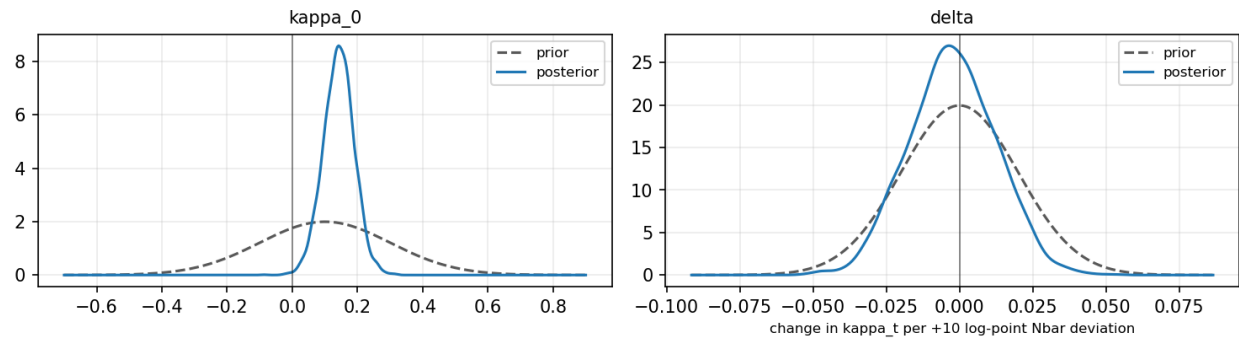
ces

Prior vs posterior — ces



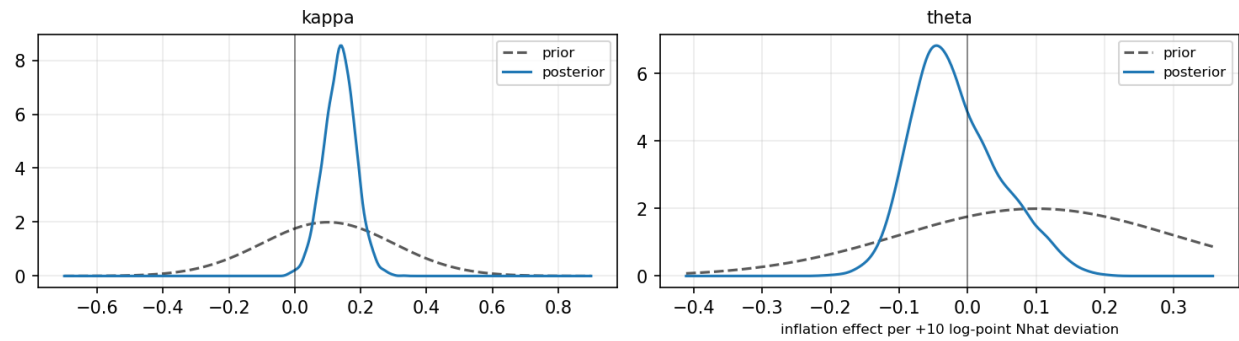
hsa_steady

Prior vs posterior — hsa_steady



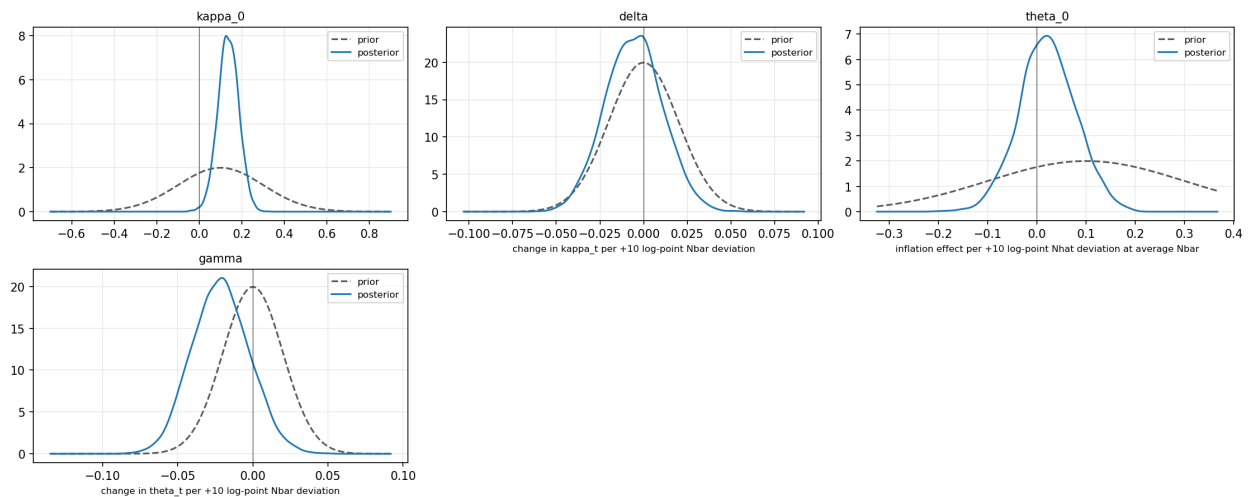
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

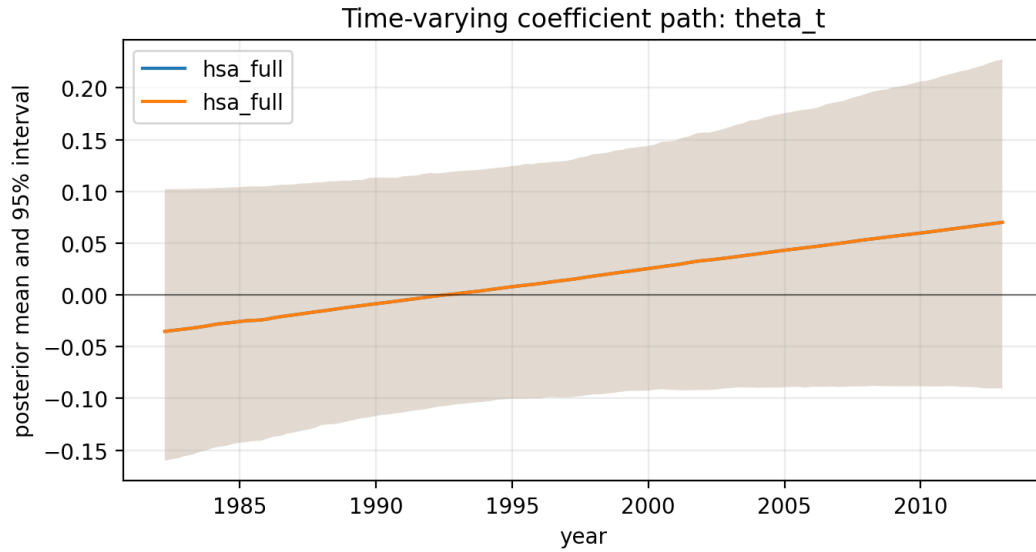
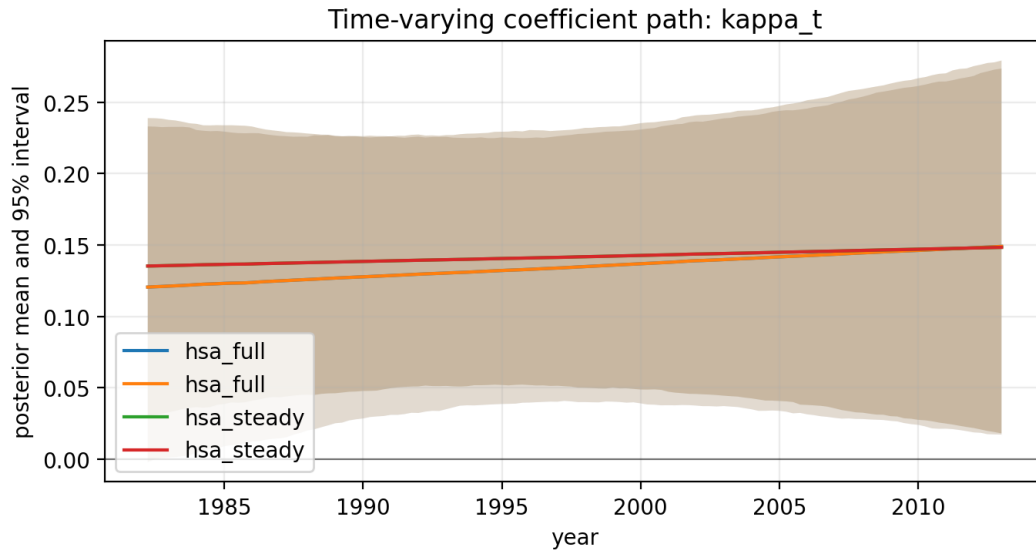
Prior vs posterior — hsa_full



6.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	0.0587
ces	kappa=0	0.1000	0.2000	0.0587
hsa_dynamic	kappa=0	0.1000	0.2000	0.1232
hsa_dynamic	theta=0	0.1000	0.2000	2.7486
hsa_dynamic	kappa=0	0.1000	0.2000	0.1232
hsa_dynamic	theta=0	0.1000	0.2000	2.7486
hsa_full	kappa_0=0	0.1000	0.2000	0.1185
hsa_full	delta=0	0.0000	0.0200	1.1700
hsa_full	theta_0=0	0.1000	0.2000	3.7373
hsa_full	gamma=0	0.0000	0.0200	0.5435
hsa_full	kappa_0=0	0.1000	0.2000	0.1185
hsa_full	delta=0	0.0000	0.0200	1.1700
hsa_full	theta_0=0	0.1000	0.2000	3.7373
hsa_full	gamma=0	0.0000	0.0200	0.5435
hsa_steady	kappa_0=0	0.1000	0.2000	0.0622
hsa_steady	delta=0	0.0000	0.0200	1.3089
hsa_steady	kappa_0=0	0.1000	0.2000	0.0622
hsa_steady	delta=0	0.0000	0.0200	1.3089

6.4 Time-Varying Kappa Path



6.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-114.0771	0.6059	-239.3121	1.0000	NaN	NaN	NaN
ces	-114.0771	0.6059	-239.3121	1.0000	NaN	NaN	NaN
hsa_dynamic	-113.6763	0.6048	-118.0203	47464376793473224383601480036067854443818841102352384.0000	NaN	2.7486	NaN
hsa_dynamic	-113.6763	0.6048	-118.0203	47464376793473224383601480036067854443818841102352384.0000	NaN	2.7486	NaN
hsa_full	-112.7260	0.5989	-120.9585	2513660899225563054745740266446276421107582639276032.0000	1.1700	NaN	0.5435
hsa_full	-112.7260	0.5989	-120.9585	2513660899225563054745740266446276421107582639276032.0000	1.1700	NaN	0.5435
hsa_steady	-114.0236	0.6055	-116.5812	200147671827111095606321135990955225929926300829483008.0000	1.3089	NaN	NaN
hsa_steady	-114.0236	0.6055	-116.5812	200147671827111095606321135990955225929926300829483008.0000	1.3089	NaN	NaN

7 output_gap_hp: annual_q4, prior=baseline, period=full

Condition. data=output_gap_hp, competition=annual_q4, prior=baseline, period=full, constraint=unrestricted (unrestricted).

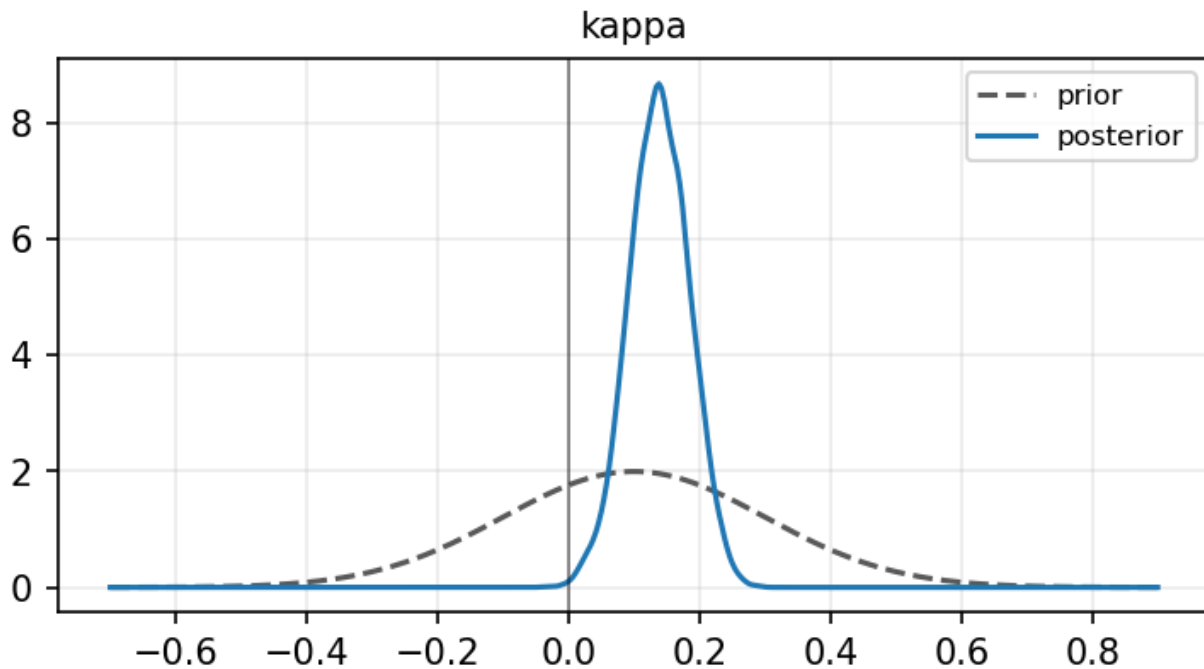
7.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.670 [0.566, 0.773]	0.665 [0.559, 0.770]	0.668 [0.567, 0.769]	0.669 [0.561, 0.779]
kappa	0.138 [0.047, 0.227]	NaN	0.143 [0.055, 0.230]	NaN
kappa_0	NaN	0.140 [0.049, 0.228]	NaN	0.140 [0.048, 0.230]
delta	NaN	-0.005 [-0.038, 0.027]	NaN	-0.005 [-0.038, 0.027]
theta	NaN	NaN	0.054 [-0.157, 0.328]	NaN
theta_0	NaN	NaN	NaN	0.049 [-0.141, 0.263]
gamma	NaN	NaN	NaN	-0.001 [-0.039, 0.036]
rho_1	NaN	0.279 [0.092, 0.463]	0.296 [0.016, 0.658]	0.279 [-0.122, 0.441]
rho_2	NaN	-0.844 [-0.967, -0.380]	-0.757 [-0.957, -0.200]	-0.852 [-0.963, -0.442]
phi_1	0.896 [0.821, 0.972]	0.895 [0.823, 0.970]	0.898 [0.826, 0.966]	0.897 [0.820, 0.969]
lambda_ez	0.134 [-0.093, 0.361]	0.130 [-0.106, 0.371]	NaN	0.133 [-0.100, 0.360]
n	NaN	-0.033 [-0.064, 0.000]	-0.030 [-0.062, 0.006]	-0.031 [-0.060, 0.001]

7.2 Prior vs Posterior by Model

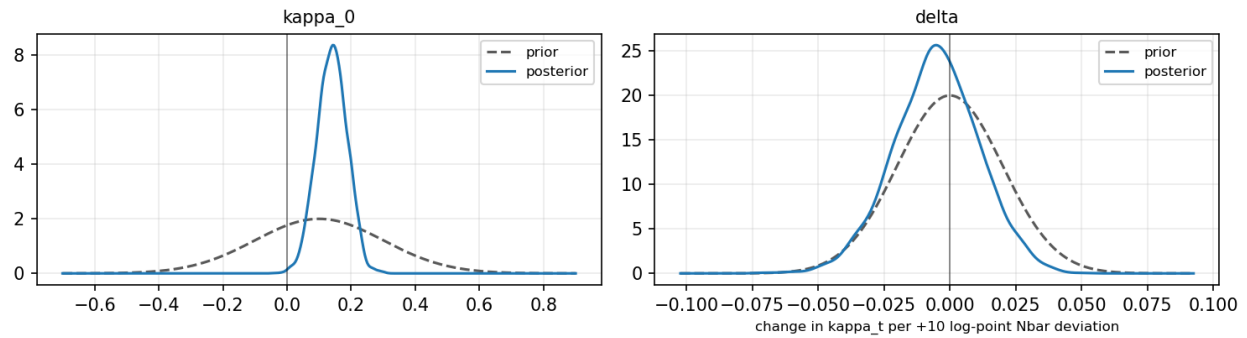
ces

Prior vs posterior — ces



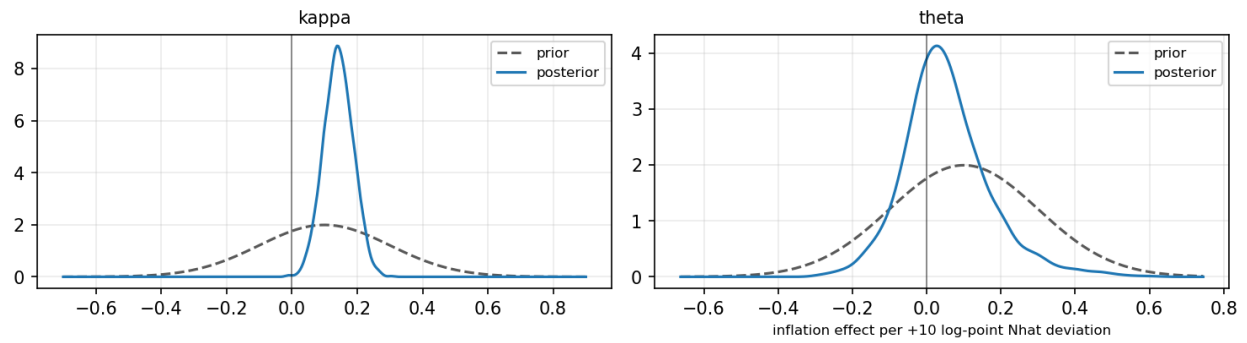
hsa_steady

Prior vs posterior — hsa_steady



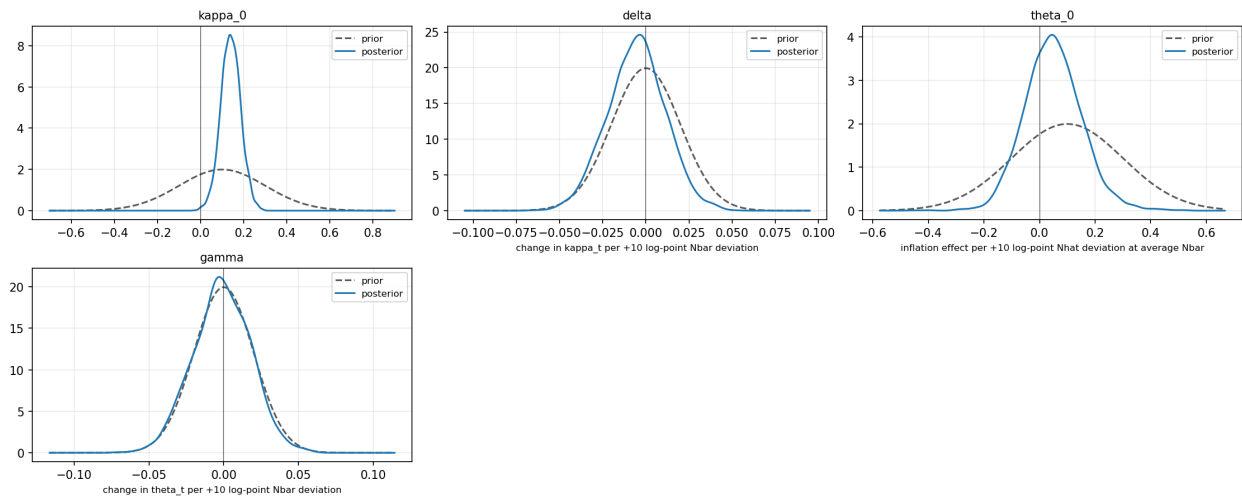
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

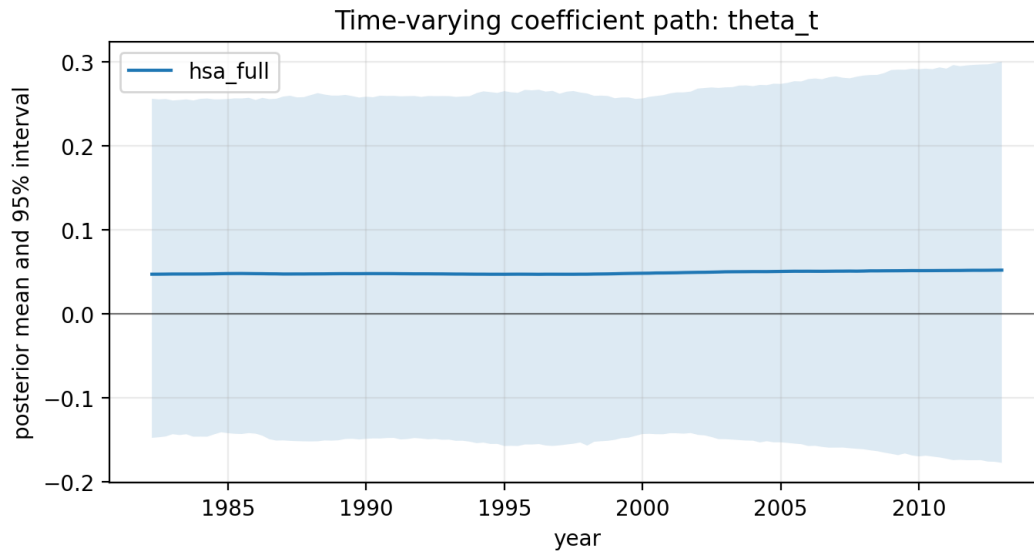
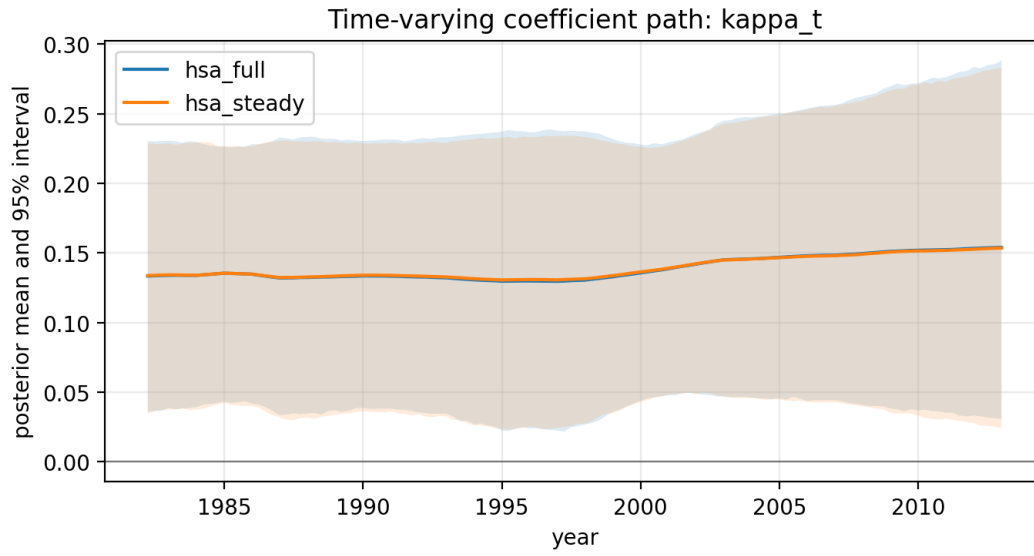
Prior vs posterior — hsa_full



7.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	0.0587
hsa_dynamic	kappa=0	0.1000	0.2000	0.0310
hsa_dynamic	theta=0	0.1000	0.2000	2.2129
hsa_full	kappa_0=0	0.1000	0.2000	0.0937
hsa_full	delta=0	0.0000	0.0200	1.1828
hsa_full	theta_0=0	0.1000	0.2000	2.0589
hsa_full	gamma=0	0.0000	0.0200	1.0388
hsa_steady	kappa_0=0	0.1000	0.2000	0.0804
hsa_steady	delta=0	0.0000	0.0200	1.1886

7.4 Time-Varying Kappa Path



7.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
os	-114.0771	0.6059	-239.3121	1.0000	NaN	NaN	NaN
hsa_dynamic	-113.8068	0.6053	1069.2866	inf	NaN	2.2129	NaN
hsa_full	-113.9562	0.6049	-120.8915	268802203956905504790576689740724874137965959315456.0000	1.1828	NaN	1.0388
hsa_steady	-113.9686	0.6051	1198.2269	inf	1.1886	NaN	NaN

8 output_gap_hp: quarterly interpolated, prior=baseline, period=full, constraint=kappa λ = 0

Condition. data=output_gap_hp, competition=quarterly_interpolated, prior=baseline, period=full, constraint=restricted_kappa (kappa λ = 0).

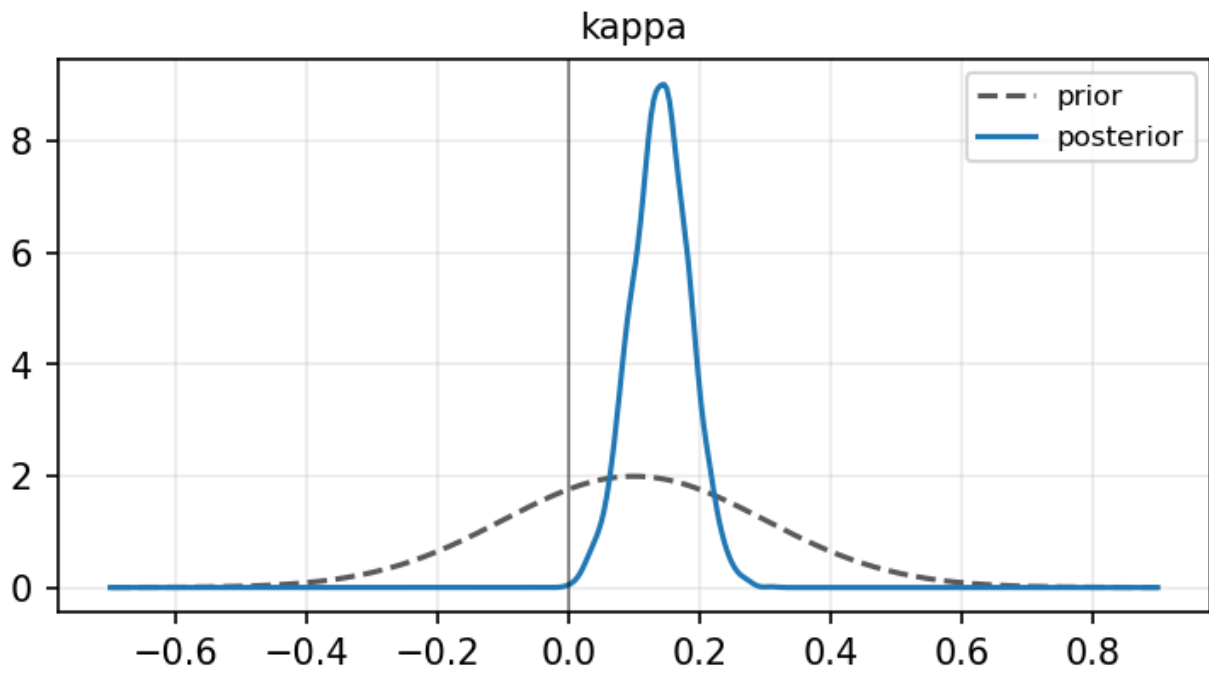
8.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.672 [0.564, 0.776]	0.664 [0.561, 0.762]	0.639 [0.523, 0.747]	0.624 [0.507, 0.739]
kappa	0.140 [0.052, 0.227]	NaN	0.141 [0.050, 0.238]	NaN
kappa_0	NaN	0.135 [0.042, 0.225]	NaN	0.139 [0.035, 0.239]
delta	NaN	-0.011 [-0.044, 0.024]	NaN	-0.011 [-0.047, 0.024]
theta	NaN	NaN	0.149 [-0.223, 0.579]	NaN
theta_0	NaN	NaN	NaN	0.220 [0.015, 0.518]
gamma	NaN	NaN	NaN	-0.003 [-0.043, 0.036]
rho_1	NaN	0.755 [0.403, 1.044]	0.887 [0.524, 1.123]	0.967 [0.701, 1.208]
rho_2	NaN	-0.018 [-0.300, 0.233]	0.005 [-0.218, 0.216]	-0.051 [-0.276, 0.177]
phi_1	0.896 [0.821, 0.972]	0.895 [0.822, 0.968]	0.898 [0.831, 0.969]	0.897 [0.825, 0.972]
lambda_ez	0.131 [-0.093, 0.358]	0.127 [-0.098, 0.351]	NaN	0.099 [-0.131, 0.326]
n	NaN	-0.020 [-0.083, 0.041]	-0.023 [-0.072, 0.025]	-0.008 [-0.061, 0.044]

8.2 Prior vs Posterior by Model

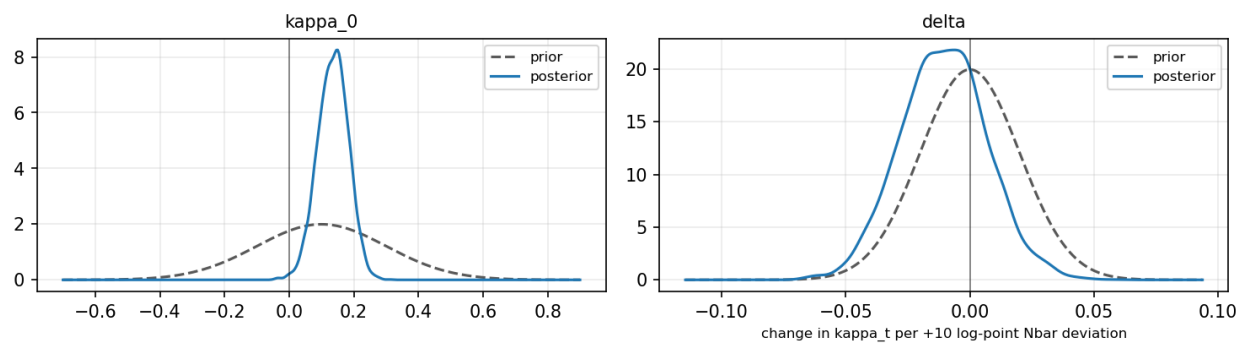
ces

Prior vs posterior — ces



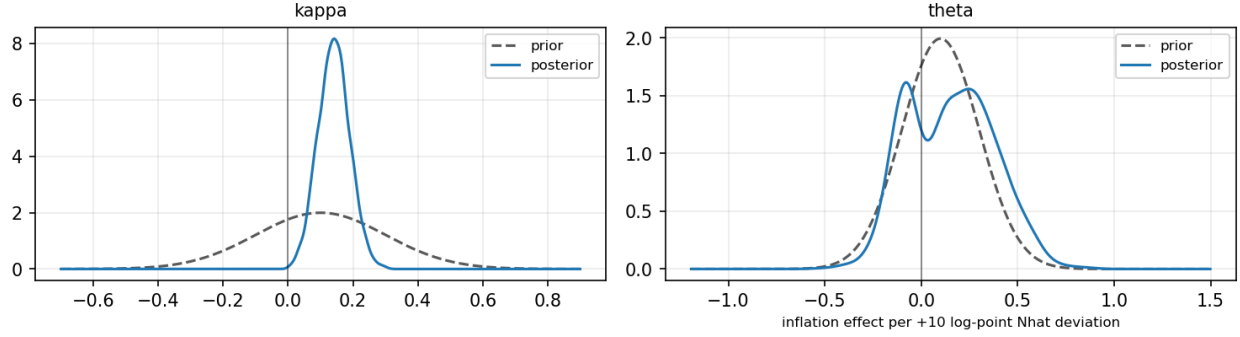
hsa_steady

Prior vs posterior — hsa_steady



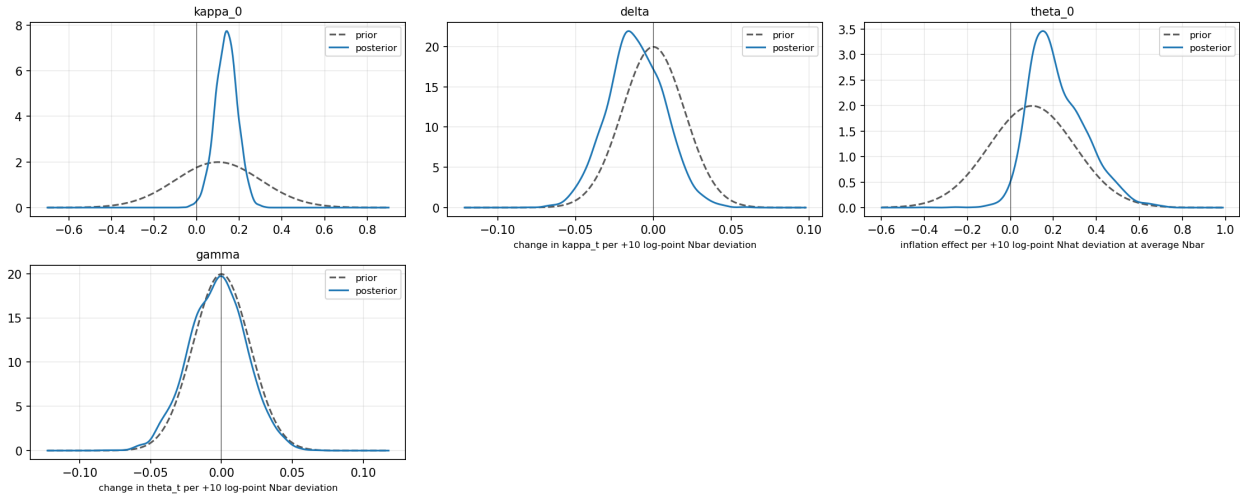
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

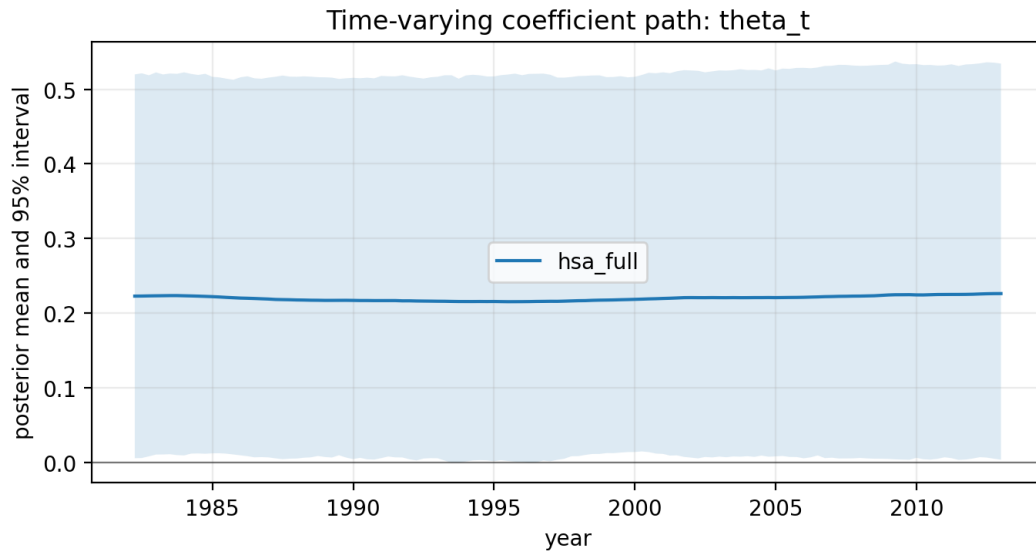
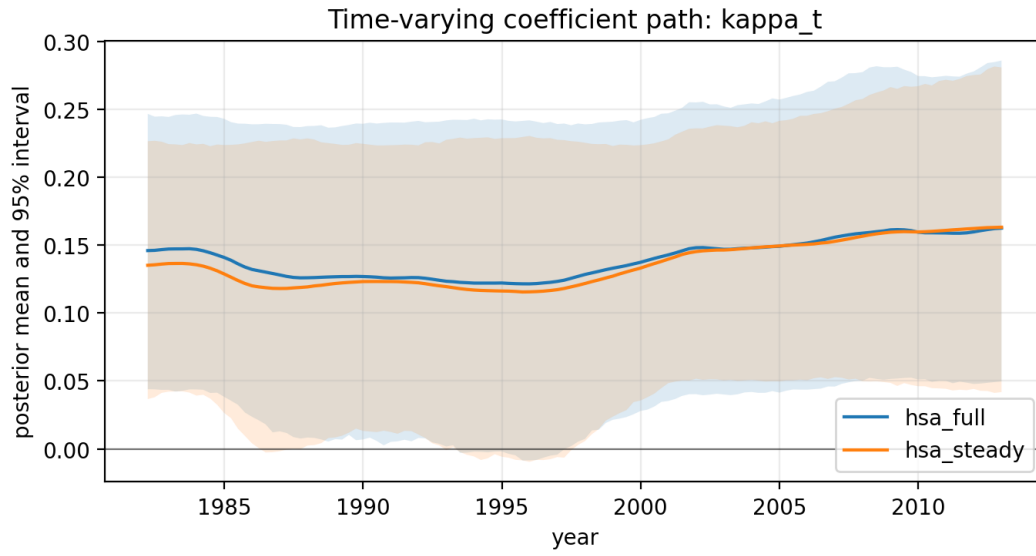
Prior vs posterior — hsa_full



8.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	0.0282
hsa_dynamic	kappa=0	0.1000	0.2000	0.0459
hsa_dynamic	theta=0	0.1000	0.2000	0.6847
hsa_full	kappa_0=0	0.1000	0.2000	0.1666
hsa_full	delta=0	0.0000	0.0200	0.8608
hsa_full	theta_0=0	0.1000	0.2000	0.2893
hsa_full	gamma=0	0.0000	0.0200	0.9883
hsa_steady	kappa_0=0	0.1000	0.2000	0.1246
hsa_steady	delta=0	0.0000	0.0200	1.0043

8.4 Time-Varying Kappa Path



8.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-114.0810	0.6058	-239.3093	1.0000	NaN	NaN	NaN
hsa_dynamic	-112.3597	0.5987	-369.9381	0.0000	NaN	0.6847	NaN
hsa_full	-105.9814	0.5657	-112.9259	7719828763192614277022234655003551898951341456618422272.0000	0.8608	NaN	0.9883
hsa_steady	-113.6351	0.6035	-348.8193	0.0000	1.0043	NaN	NaN

9 output_gap_hp: quarterly interpolated, prior=baseline, period=full, constraint=kappa t restricted

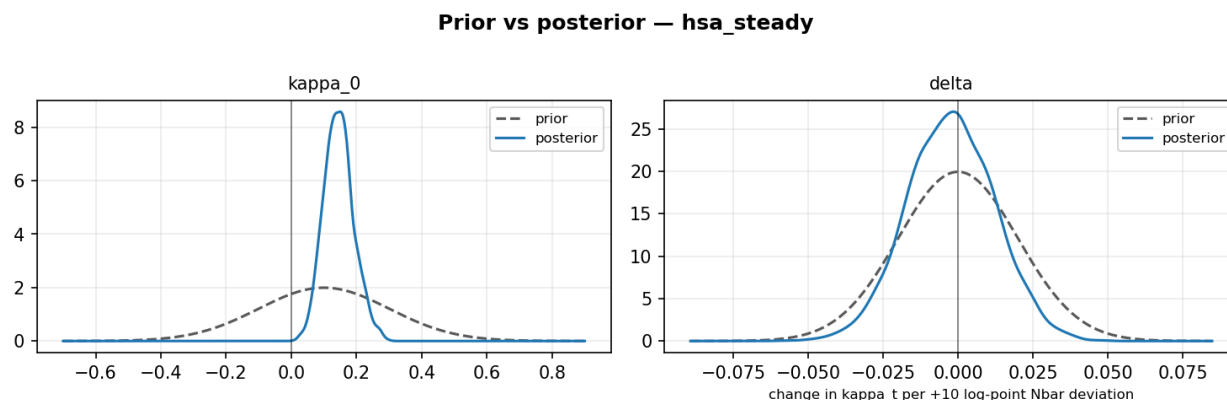
Condition. data=output_gap_hp, competition=quarterly_interpolated, prior=baseline, period=full, constraint=restricted_kappa_t (kappa t restricted).

9.1 Coefficient Table

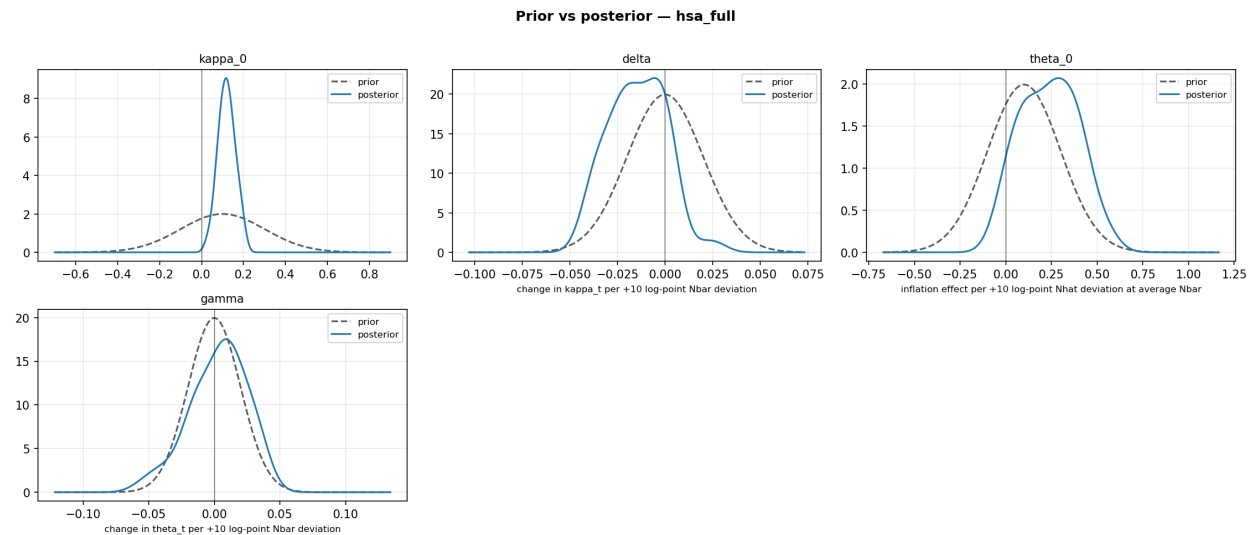
	hsa_steady	hsa_full
parameter		
alpha	0.667 [0.563, 0.773]	0.658 [0.582, 0.737]
kappa_0	0.145 [0.061, 0.241]	0.120 [0.049, 0.198]
delta	-0.002 [-0.031, 0.026]	-0.015 [-0.038, 0.004]
theta_0	NaN	0.238 [-0.002, 0.515]
gamma	NaN	0.003 [-0.041, 0.033]
rho_1	1.806 [1.708, 1.896]	0.854 [0.524, 1.092]
rho_2	-0.820 [-0.907, -0.725]	-0.113 [-0.336, 0.064]
phi_1	0.898 [0.823, 0.971]	0.909 [0.821, 0.977]
lambda_ez	0.125 [-0.103, 0.349]	0.122 [-0.073, 0.290]
n	-0.044 [-0.063, -0.020]	-0.016 [-0.054, 0.016]

9.2 Prior vs Posterior by Model

hsa_steady



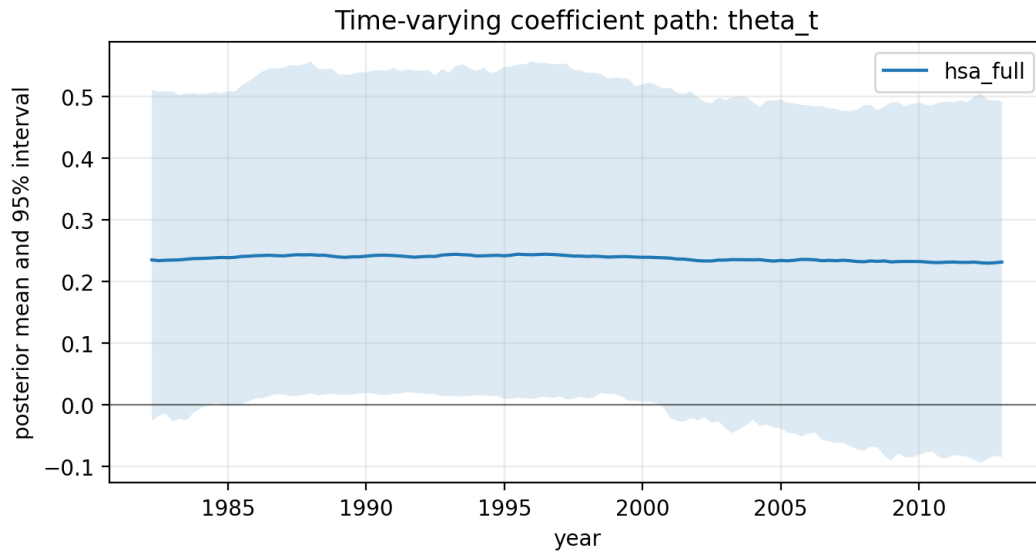
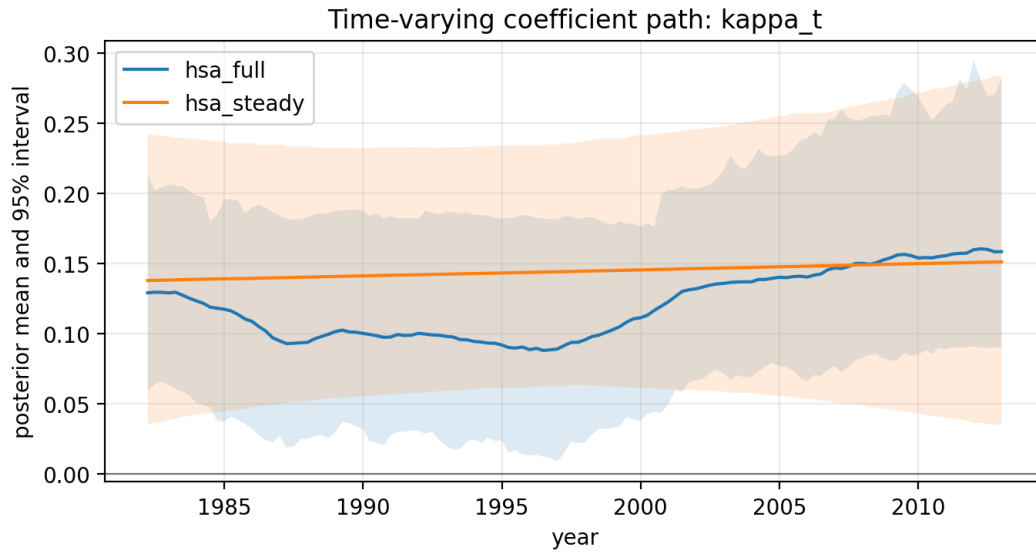
hsa_full



9.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
hsa_full	$\kappa_0=0$	0.1000	0.2000	0.0965
hsa_full	$\delta=0$	0.0000	0.0200	1.0007
hsa_full	$\theta_0=0$	0.1000	0.2000	0.6485
hsa_full	$\gamma=0$	0.0000	0.0200	0.7990
hsa_steady	$\kappa_0=0$	0.1000	0.2000	0.0058
hsa_steady	$\delta=0$	0.0000	0.0200	1.3417

9.4 Time-Varying Kappa Path



9.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
hsa_full	-110.4495	0.5888	-117.4255	1.0000	1.0007	NaN	0.7990
hsa_steady	-114.0117	0.6053	-116.5410	2.4216	1.3417	NaN	NaN

10 output_gap_hp: quarterly_interpolated, prior=weak, period=full

Condition. data=output_gap_hp, competition=quarterly_interpolated, prior=weak, period=full, constraint=unrestricted (unrestricted).

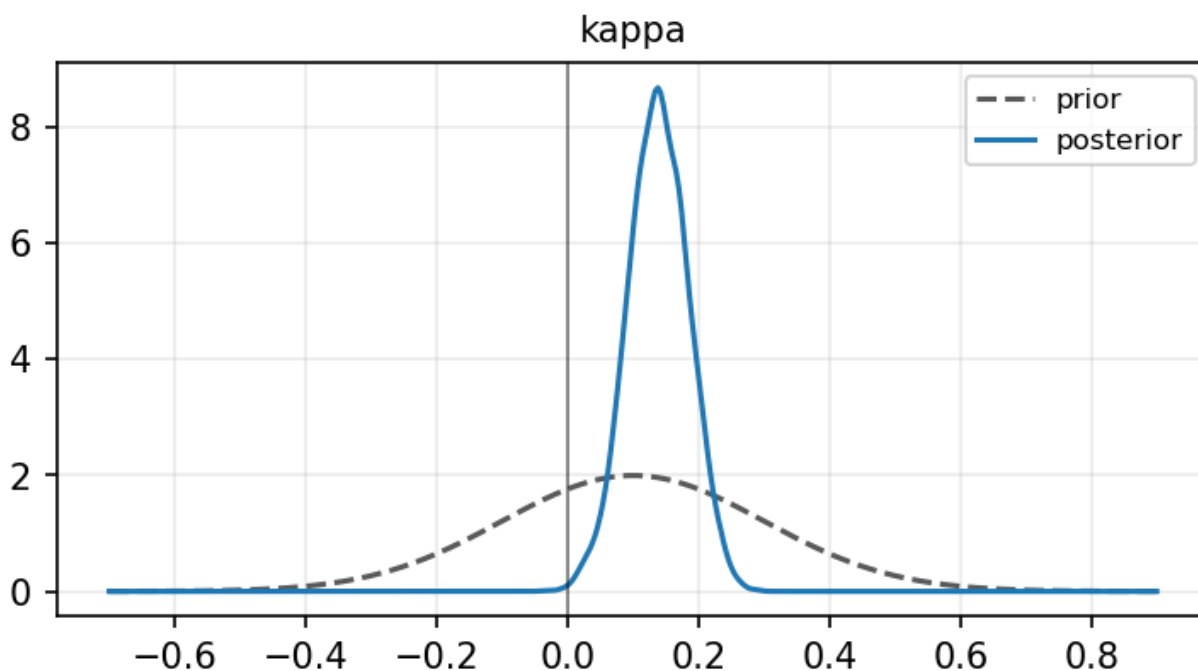
10.1 Coefficient Table

	ces	hsa_steady	hsa_dynamic	hsa_full
parameter				
alpha	0.670 [0.566, 0.773]	0.666 [0.564, 0.768]	0.662 [0.555, 0.763]	0.660 [0.548, 0.764]
kappa	0.138 [0.047, 0.227]	NaN	0.137 [0.042, 0.232]	NaN
kappa_0	NaN	0.142 [0.048, 0.233]	NaN	0.136 [0.040, 0.229]
delta	NaN	-0.002 [-0.031, 0.026]	NaN	-0.006 [-0.037, 0.026]
theta	NaN	NaN	-0.017 [-0.122, 0.123]	NaN
theta_0	NaN	NaN	NaN	0.022 [-0.091, 0.135]
gamma	NaN	NaN	NaN	-0.021 [-0.057, 0.017]
rho_1	NaN	1.805 [1.703, 1.896]	1.781 [1.667, 1.886]	1.798 [1.687, 1.891]
rho_2	NaN	-0.818 [-0.905, -0.721]	-0.794 [-0.894, -0.685]	-0.813 [-0.903, -0.705]
phi_1	0.896 [0.821, 0.972]	0.898 [0.822, 0.973]	0.898 [0.824, 0.968]	0.896 [0.821, 0.969]
lambda_ez	0.134 [-0.093, 0.361]	0.127 [-0.100, 0.356]	NaN	0.132 [-0.090, 0.357]
n	NaN	-0.044 [-0.063, -0.020]	-0.044 [-0.069, 0.001]	-0.039 [-0.050, -0.029]

10.2 Prior vs Posterior by Model

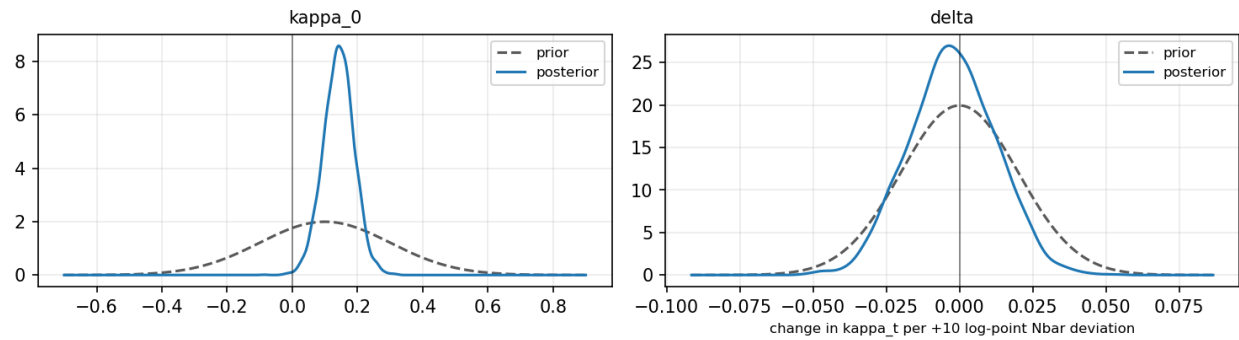
ces

Prior vs posterior — ces



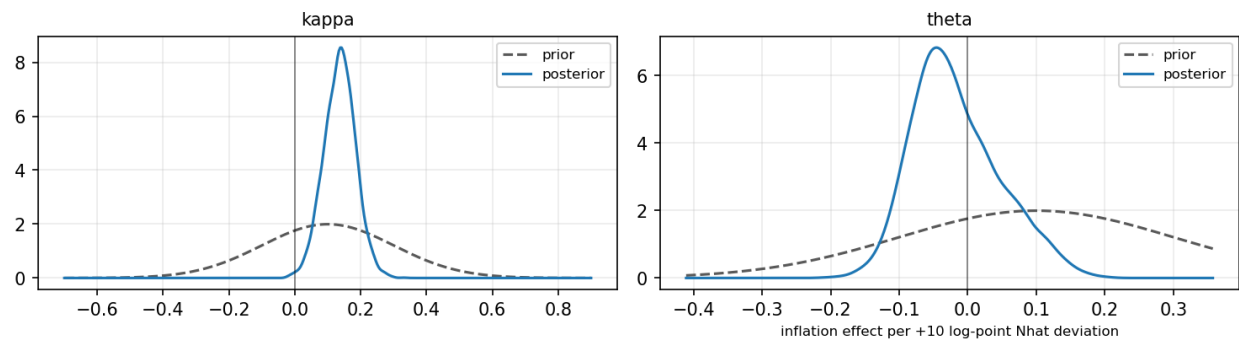
hsa_steady

Prior vs posterior — hsa_steady



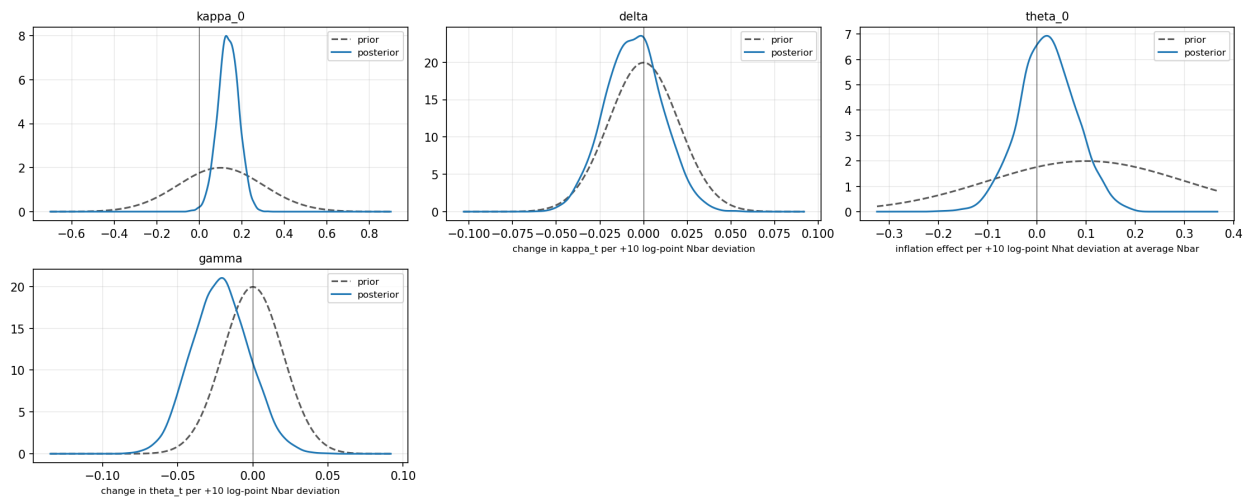
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

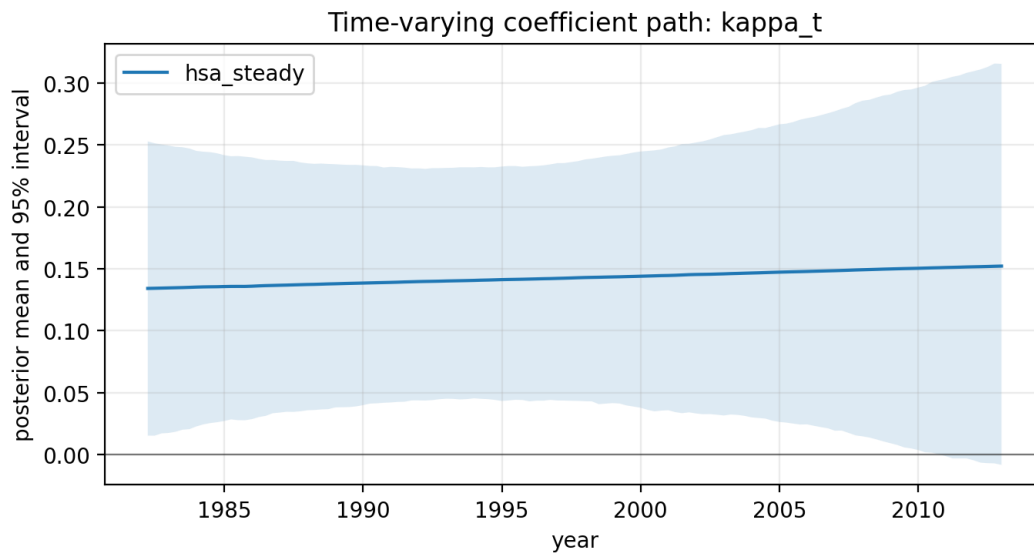
Prior vs posterior — hsa_full



10.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
hsa_steady	kappa_0=0	0.1000	0.5000	0.4479
hsa_steady	delta=0	0.0000	0.1000	3.2406

10.4 Time-Varying Kappa Path



No theta_t path for this block.

10.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
hsa_steady	-114.0277	0.6060	-108.7864	1.0000	3.2406	NaN	NaN

11 output_gap_hp: quarterly_interpolated, prior=tight, period=full

Condition. data=output_gap_hp, competition=quarterly_interpolated, prior=tight, period=full, constraint=unrestricted (unrestricted).

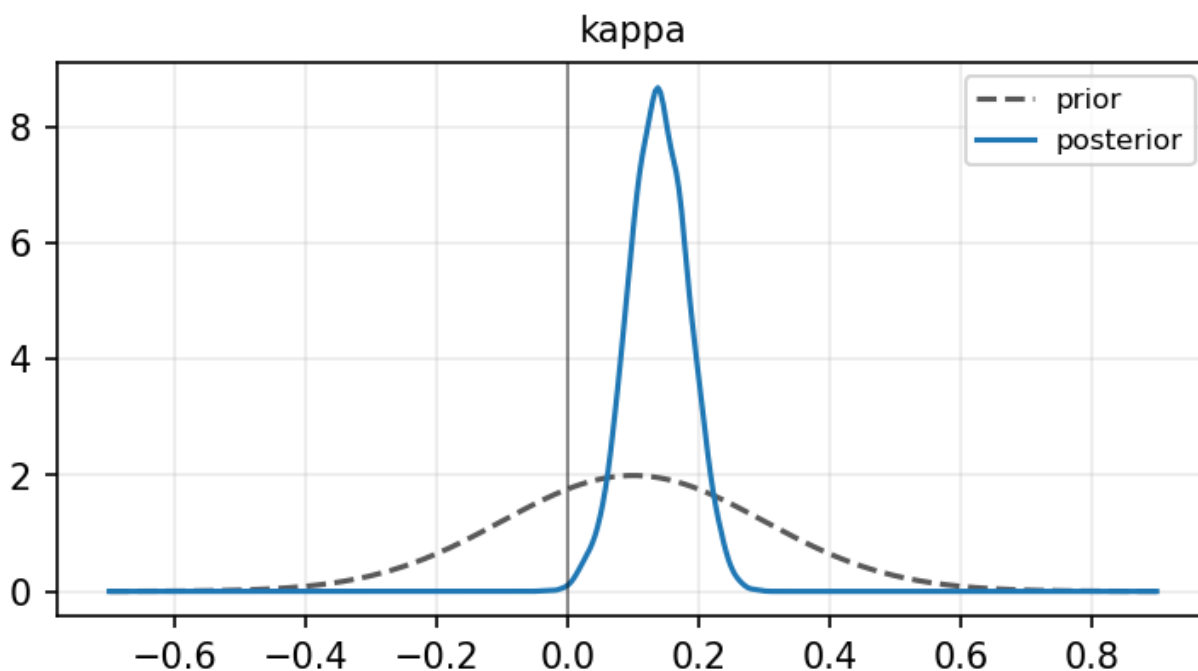
11.1 Coefficient Table

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.670 [0.566, 0.773]	0.666 [0.564, 0.768]	0.662 [0.555, 0.763]	0.660 [0.548, 0.764]
kappa	0.138 [0.047, 0.227]	NaN	0.137 [0.042, 0.232]	NaN
kappa_0	NaN	0.142 [0.048, 0.233]	NaN	0.136 [0.040, 0.229]
delta	NaN	-0.002 [-0.031, 0.026]	NaN	-0.006 [-0.037, 0.026]
theta	NaN	NaN	-0.017 [-0.122, 0.123]	NaN
theta_0	NaN	NaN	NaN	0.022 [-0.091, 0.135]
gamma	NaN	NaN	NaN	-0.021 [-0.057, 0.017]
rho_1	NaN	1.805 [1.703, 1.896]	1.781 [1.667, 1.886]	1.798 [1.687, 1.891]
rho_2	NaN	-0.818 [-0.905, -0.721]	-0.794 [-0.894, -0.685]	-0.813 [-0.903, -0.705]
phi_1	0.896 [0.821, 0.972]	0.898 [0.822, 0.973]	0.898 [0.824, 0.968]	0.896 [0.821, 0.969]
lambda_ez	0.134 [-0.093, 0.361]	0.127 [-0.100, 0.356]	NaN	0.132 [-0.090, 0.357]
n	NaN	-0.044 [-0.063, -0.020]	-0.044 [-0.069, 0.001]	-0.039 [-0.050, -0.029]

11.2 Prior vs Posterior by Model

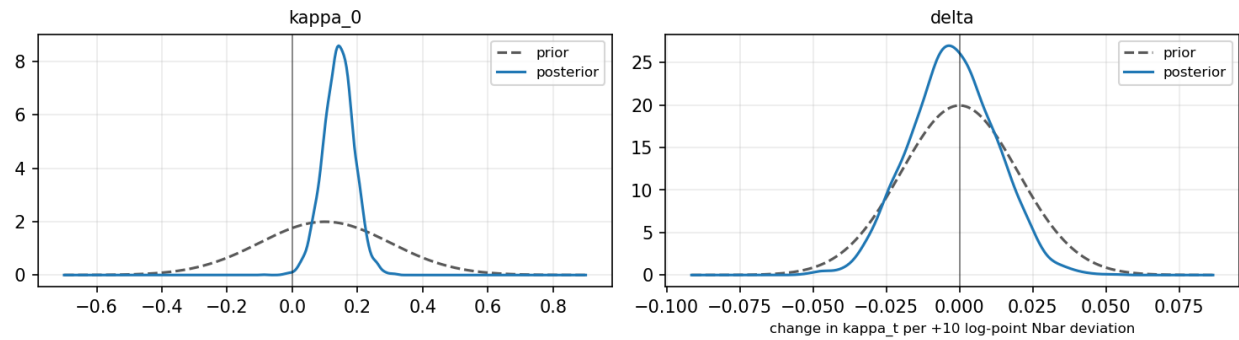
ces

Prior vs posterior — ces



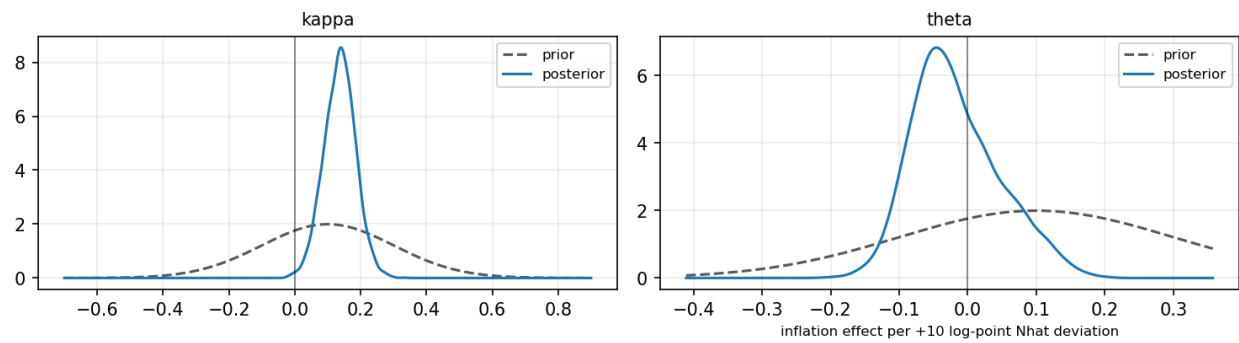
hsa_steady

Prior vs posterior — hsa_steady



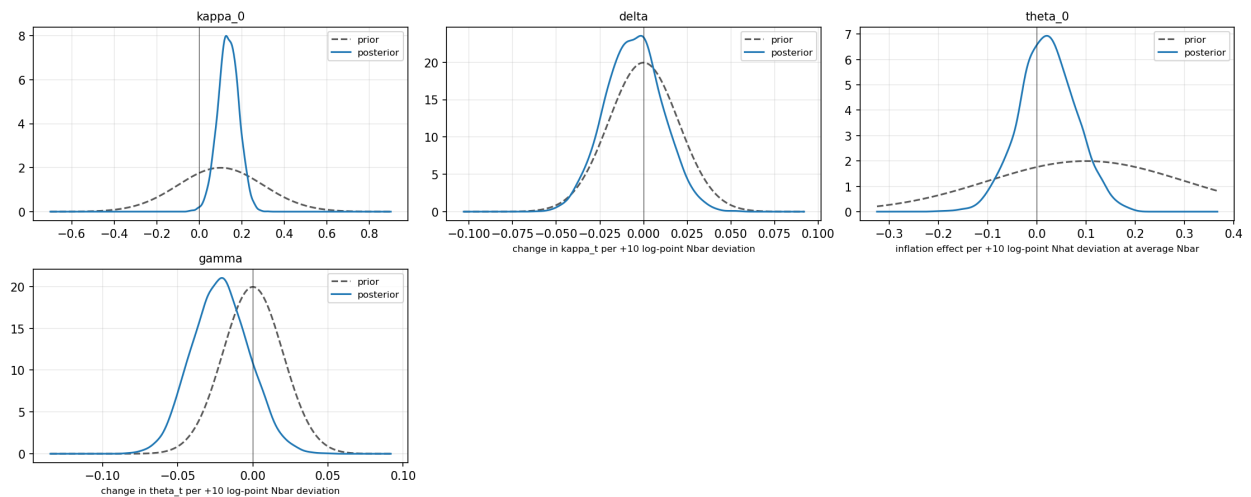
hsa_dynamic

Prior vs posterior — hsa_dynamic



hsa_full

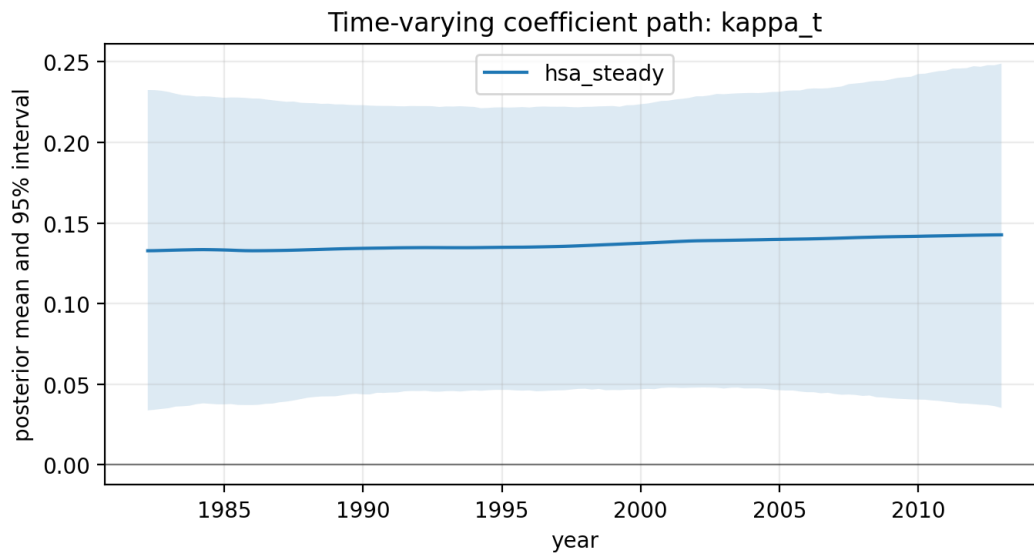
Prior vs posterior — hsa_full



11.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
hsa_steady	kappa_0=0	0.1000	0.1000	0.0288
hsa_steady	delta=0	0.0000	0.0100	1.0786

11.4 Time-Varying Kappa Path



No theta_t path for this block.

11.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
hsa_steady	-114.3819	0.6057	-179.8429	1.0000	1.0786	NaN	NaN

12 Competition Decomposition

For annual-Q4 mixed-frequency HSA runs, Table 2 reports selected quarters of the posterior decomposition $N_t = \bar{N}_t + \hat{N}_t$. The full quarterly decomposition is written to the corresponding CSV artifact.

Table 2: Posterior decomposition of competition N_t

quarter	model	N_total_mean	N_total.q05	N_total.q50	N_total.q95	Nbar_mean	Nhat_mean
1982Q1	hsa_full	3.3698	-0.9323	3.5249	6.6134	1.3548	2.0150
1984Q1	hsa_full	1.0762	-0.3553	1.1143	2.3915	1.1297	-0.0535
1986Q1	hsa_full	0.2163	-0.7233	0.1456	1.4097	1.0991	-0.8827
1988Q2	hsa_full	0.9158	-0.1322	0.8718	2.0262	1.3538	-0.4381
1990Q2	hsa_full	1.1644	0.4184	1.1829	1.8877	1.1971	-0.0326
1992Q2	hsa_full	1.5414	0.6367	1.5719	2.4057	1.3752	0.1662
1994Q2	hsa_full	1.7579	0.9305	1.8305	2.4563	1.7747	-0.0168
1996Q2	hsa_full	1.6381	0.7519	1.7195	2.3315	1.8531	-0.2150
1998Q3	hsa_full	1.2660	0.5943	1.3117	1.8156	1.3910	-0.1251
2000Q3	hsa_full	0.1330	-0.4634	0.1630	0.6258	0.3502	-0.2172
2002Q3	hsa_full	-0.8689	-1.3823	-0.8573	-0.3534	-0.8158	-0.0531
2004Q3	hsa_full	-1.1076	-1.6202	-1.1111	-0.5703	-1.2239	0.1163
2006Q3	hsa_full	-1.4589	-1.9383	-1.4528	-1.0020	-1.5108	0.0519
2008Q4	hsa_full	-2.0161	-2.1244	-2.0177	-1.8991	-2.0161	0.0000
2010Q4	hsa_full	-2.3045	-2.4198	-2.3038	-2.1896	-2.2803	-0.0242
2012Q4	hsa_full	-2.6273	-2.7473	-2.6269	-2.5138	-2.6054	-0.0219

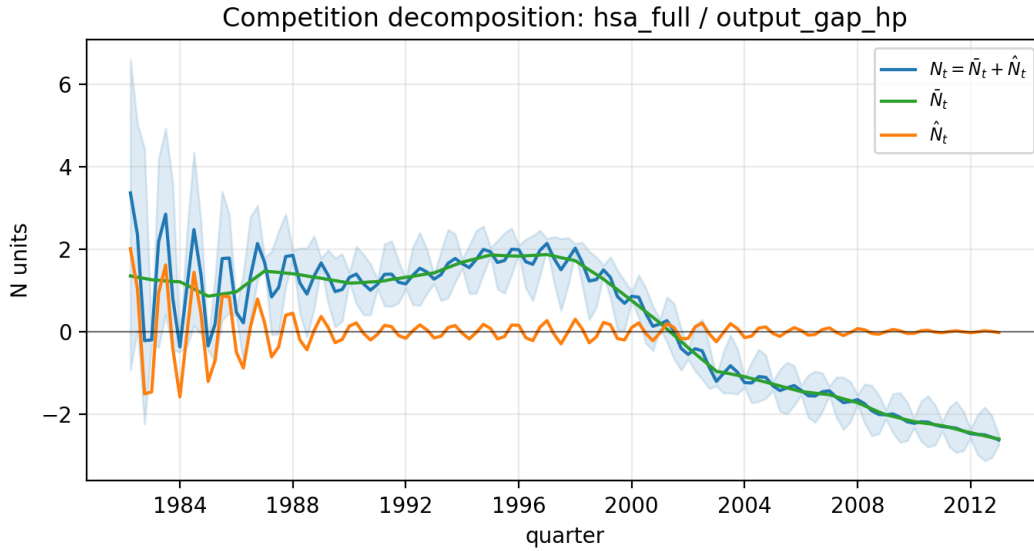


Figure 1: Posterior mean decomposition of competition

13 Notes

WAIC and LOO are not used as primary criteria because the models contain latent states. Chib marginal likelihoods are reported only when the legacy ordinate calculation can be matched to the sampler units and supplied data. For the full HSA model, the reported Chib quantity is conditional

on posterior mean latent states because the observation equation is nonlinear in $\hat{N}_t$ and $\bar{N}_t$.